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Electrical power outlet having flexible cover

Abstract

An electrical power outlet includes a housing, a faceplate having a plurality of slots configured to receive a plurality of prongs on an electrical plug, a plurality of electrical terminals configured to receive the plurality of prongs to provide electrical connections and a flexible cover disposed between the plurality of slots and the plurality of terminals. The flexible cover having a plurality of slits configured to receive the plurality of prongs in an open position and for covering access to the plurality of terminals in a closed position.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure generally relates to electrical power outlets, and more particularly relates to an electrical power outlet having a tamper resistant layer, particularly for use in a vehicle, for example.

BACKGROUND OF THE DISCLOSURE

(2) Motor vehicles are increasingly equipped with electrical power outlets for powering and

charging electronic devices. Both direct current (DC) and alternating current (AC) power outlets can be found on motor vehicles. DC power from the vehicle battery may be converted with an inverter to AC power. It may be desirable to provide for a power outlet that is tamper resistant.

SUMMARY OF THE DISCLOSURE

(3) According to a first aspect of the present disclosure, an electrical power outlet is provided. The electrical power outlet includes a housing, a faceplate having a plurality of slots configured to receive a plurality of prongs on an electrical plug, a plurality of electrical terminals configured to receive the plurality of prongs to provide electrical connections, and a flexible cover disposed between the plurality of slots and the plurality of terminals, the flexible cover having a plurality of slits configured to receive the plurality of prongs in an open position and for covering access to the plurality of terminals in a closed position.

(4) Embodiments of the first aspect of the present disclosure can include any one or a combination of the following features: a switch coupled to at least one of the plurality of electrical terminals to detect the connection of a first prong to a first electrical terminal of the plurality of electrical terminals; the switch activates electrical power supplied to the plurality of prongs when the switch detects connection of the first prong to the first electrical terminal; a circuit board, wherein the plurality of electrical terminals are mounted on the circuit board and the switch is aligned to be engaged with the first electrical terminal; the switch is coupled to the circuit board; a light source configured to illuminate at least a region proximate the faceplate when the switch detects connection of the first prong to the first electrical terminal; the light source comprises a light transmissive optic element forming a ring surrounding the faceplate; the light source comprises a light-emitting diode (LED); the cover comprises a membrane of flexible material; the housing comprises a first bezel configured to connect to a second bezel; and the electrical power outlet is located on a motor vehicle.

(5) According to a second aspect of the present disclosure, an electrical power outlet is provided. The electrical power outlet includes a housing, a faceplate having a plurality of slots configured to receive a plurality of prongs on an electrical plug, and a plurality of electrical terminals configured to receive the plurality of prongs to provide electrical connections. The electrical power outlet also includes a flexible cover disposed between the plurality of slots and the plurality of terminals, the flexible cover having a plurality of slits configured to receive the plurality of prongs in an open position and for covering access to the plurality of terminals in a closed position, and a switch coupled to a first electrical terminal of the plurality of electrical terminals to detect the connection of a first prong to the first electrical terminals, wherein the switch activates electrical power supplied to the plurality of prongs when the switch detects connection of the first prong to the first electrical terminal. The electrical power outlet further includes a circuit board, wherein the plurality of electrical terminals are mounted on the circuit board and the switch is aligned to be engaged with the first electrical terminal, wherein the switch is coupled to the circuit board, and a light source configured to illuminate a region proximate the faceplate when the switch detects connection of the first prong to the first electrical terminal.

(6) According to a third aspect of the present disclosure, a vehicle is provided including an electrical power source, power lines receiving electrical power from the electrical power source, and an electrical power outlet coupled to the power lines. The electrical power outlet including a housing, a faceplate having a plurality of slots configured to receive a plurality of prongs on an electrical plug, a plurality of electrical terminals configured to receive the plurality of prongs to provide electrical connections, and a flexible cover disposed between the plurality of slots and the plurality of terminals, the flexible cover having a plurality of slits configured to receive the plurality of prongs in an open position and for covering access to the plurality of terminals in a closed position.

(7) Embodiments of the third aspect of the present disclosure can include any one or a combination of the following features: a switch coupled to a first electrical terminal of the plurality of terminals

to detect the connection of a prong in the first electrical terminal, wherein the switch activates electrical power supplied to the plurality of prongs when the switch detects connection of the first prong to the first electrical terminal; a circuit board, wherein the plurality of electrical terminals and the switch are mounted on the circuit board and the switch is aligned to be engaged with the at the first electrical terminal on the circuit board; a light source configured to illuminate at least a region proximate the faceplate when the switch detects connection of the first prong to the first electrical terminal or change color according to a status of the plug; the light source comprises a light transmissive optic element forming a ring surrounding the faceplate; the light source comprises a light-emitting diode (LED); the cover comprises a membrane of flexible material; and the plurality of slits includes I-shaped slits.
(8) These and other features, advantages, and objects of the present disclosure will be further understood and appreciated by those skilled in the art by reference to the following specification, claims, and appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) In the drawings:
(2) FIG. **1** is a perspective view of a front portion of an interior of a motor vehicle having an electrical power outlet, according to one embodiment;
(3) FIG. **2** is an enlarged front view of the electrical power outlet illustrated in FIG. **1**, according to one embodiment;
(4) FIG. **3** is an exploded front perspective view of the electrical power outlet illustrated in FIG. **2**;
(5) FIG. **4** is a cross-sectional view taken through line IV-IV of FIG. **3**;
(6) FIG. **5** is an exploded rear perspective view of a portion of the electrical power outlet;
(7) FIG. **6** is a cross-sectional perspective view of the electrical power outlet taken through line VI-VI of FIG. **1** with an electrical plug of a device disconnected therefrom;
(8) FIG. **6**A is a cross-sectional view taken through line VI-VI of FIG. **1** illustrating an electrical plug of for an electronic powered device inserted in the electrical power outlet;
(9) FIG. **7** is a front perspective view of an electrical power outlet having a light output ring, according to another embodiment;
(10) FIG. **8** is an exploded front perspective view of the electrical power outlet illustrated in FIG. **7**;
(11) FIG. **9** is a cross-sectional view taken through line IX-IX of FIG. **8**;
(12) FIG. **10** is a rear exploded view of a portion of the electrical power outlet shown in FIG. **7**;
(13) FIG. **11** is an exploded rear perspective view of the electrical power outlet further having clips positioned proximate to slots, according to a further embodiment; and
(14) FIG. **12** is a cross-sectional view taken through line XII-XII of FIG. **11**.
DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS
(15) Reference will now be made in detail to the present preferred embodiments of the disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numerals will be used throughout the drawings to refer to the same or like parts. In the drawings, the depicted structural elements are not to scale and certain components are enlarged relative to the other components for purposes of emphasis and understanding.
(16) As required, detailed embodiments of the present disclosure are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the disclosure that may be embodied in various and alternative forms. The figures are not necessarily to a detailed design; some schematics may be exaggerated or minimized to show function overview. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the present

disclosure.
(17) For purposes of description herein, the terms “upper,” “lower,” “right,” “left,” “rear,” “front,” “vertical,” “horizontal,” and derivatives thereof shall relate to the concepts as oriented in FIG. **1**. However, it is to be understood that the concepts may assume various alternative orientations, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.
(18) The present illustrated embodiments reside primarily in combinations of method steps and apparatus components related to an electrical power outlet and a vehicle having an electrical power outlet. Accordingly, the apparatus components and method steps have been represented, where appropriate, by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present disclosure so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein. Further, like numerals in the description and drawings represent like elements.
(19) As used herein, the term “and/or,” when used in a list of two or more items, means that any one of the listed items can be employed by itself, or any combination of two or more of the listed items, can be employed. For example, if a composition is described as containing components A, B, and/or C, the composition can contain A alone; B alone; C alone; A and B in combination; A and C in combination; B and C in combination; or A, B, and C in combination.
(20) In this document, relational terms, such as first and second, top and bottom, and the like, are used solely to distinguish one entity or action from another entity or action, without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises . . . a” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element.
(21) As used herein, the term “about” means that amounts, sizes, formulations, parameters, and other quantities and characteristics are not and need not be exact, but may be approximate and/or larger or smaller, as desired, reflecting tolerances, conversion factors, rounding off, measurement error and the like, and other factors known to those of skill in the art. When the term “about” is used in describing a value or an end-point of a range, the disclosure should be understood to include the specific value or end-point referred to. Whether or not a numerical value or end-point of a range in the specification recites “about,” the numerical value or end-point of a range is intended to include two embodiments: one modified by “about,” and one not modified by “about.” It will be further understood that the end-points of each of the ranges are significant both in relation to the other end-point, and independently of the other end-point.
(22) The terms “substantial,” “substantially,” and variations thereof as used herein are intended to note that a described feature is equal or approximately equal to a value or description. For example, a “substantially planar” surface is intended to denote a surface that is planar or approximately planar. Moreover, “substantially” is intended to denote that two values are equal or approximately equal. In some embodiments, “substantially” may denote values within about 10% of each other, such as within about 5% of each other, or within about 2% of each other.
(23) As used herein the terms “the,” “a,” or “an,” mean “at least one,” and should not be limited to “only one” unless explicitly indicated to the contrary. Thus, for example, reference to “a component” includes embodiments having two or more such components unless the context clearly

indicates otherwise.

(24) Referring to FIG. **1**, a motor vehicle **10** is generally illustrated configured with passenger seating for transporting one or more passengers including a driver of the vehicle **10** and one or more passengers, according to one example. The motor vehicle **10** has a vehicle body that generally defines a cabin interior **12**. The cabin interior **12** may include various features and trim components. The cabin interior **12** is generally shown having an arrangement of passenger seats including a front row of seating having a pair of seats **14** generally spaced apart from one another, according to one example. Located but not restricted in the space between the pair of seats **14** in the front row is a center console **16**, which may provide storage and an armrest for the driver and passenger seated in seats **14**.

(25) It should be appreciated that the motor vehicle **10** may be a wheeled motor vehicle, such as a car, truck, SUV, bus, van or other vehicle. According to other examples, the motor vehicle **10** may be an aircraft, a boat, or other motor vehicle. The motor vehicle **10** may have a motor, such as an internal combustion engine or an electrical motor or both, according to a few examples. The motor vehicle **10** has an electrical power source **18**, which is shown as a battery. It should be appreciated that one or more batteries supplying electrical power may be provided on the motor vehicle **10**. The electrical power may be in the form of a direct current (DC) voltage such as a 12-volt battery. It should also be appreciated that the motor vehicle **10** may include inverter circuitry **98** for converting the DC voltage to an alternating current (AC) voltage and power lines **90**A-**90**C for supplying the AC electrical power output to one or more electrical power outlets.

(26) The motor vehicle **10** is shown equipped with one or more electrical power outlets **20** for supplying the AC electrical power to one or more plug-in devices. In the embodiment shown, a pair of electrical power outlets **20** are shown located on the rear facing wall of the rear end of the center console **16** and are made available for passengers generally seated rearward of the front row of seating to plug-in an electrical plug **100** associated with one or more electrically-powered devices such as a phone, a computer, a speaker or any electrically-powered devices. It should further be appreciated that one or more electrical power outlets **20** may be located elsewhere on the motor vehicle **10** such as forward of the center console, within the center console **16** or within other areas that are accessible to one or more passengers in the motor vehicle **10** to supply power to one or more electrical powered devices. It should further be appreciated that the electrical power outlet **20** as shown and described herein may be used on a motor vehicle or off a vehicle, such as within a home environment, according to other embodiments.

(27) The electrical power outlet **20** is illustrated in FIGS. **2**-**6**A configured as a three-prong receptacle, according to a first embodiment. As seen in FIG. **2**, the electrical power outlet **20** has a housing **22** which is generally mounted in an opening formed in a wall of the center console **16**. The housing **22** has a faceplate **24** on the front end thereof which generally defines a region having a plurality of receptacle outlet slots configured in a shape, size and arrangement to receive a plurality of prongs, such as two or three prongs, on an electrical plug **100**. In the embodiment shown, the faceplate **24** has three receptacle outlet slots **30**A-**30**C. The first and second receptacle outlet slots **30**A and **30**B generally include an elongated portion for receiving aligned positive and negative prongs of the electrical plug **100** which, in turn, connect to positive and negative terminals for receiving positive and negative AC electrical power, according to one example. The positive AC electrical power may be referred to as the hot signal and the negative AC electrical power may be referred to as the neutral signal. The third receptacle outlet slot **30**C is located below the first and second slots **30**A and **30**C and is configured to matingly engage a ground prong on the electrical plug **100** to provide a ground electrical connection. It should be appreciated that some electrical plugs may include a ground prong and other electrical plugs may not include a ground prong. The arrangement of the three slots **30**A-**30**C shown is consistent with a North American style electrical plug, according to one example. However, it should be appreciated that the slots **30**A-**30**C may otherwise be configured according to other arrangements, such as a European style or an Asian

Pacific style electrical outlet, for example.

(28) Disposed proximate to the faceplate **24** and configured around a periphery of the faceplate **24** is a light ring area **26** which is generally formed by an open region that receives and illuminates light from a light source **48** emitted into a receiver **54** on the back side of the faceplate **24**. The light ring area **26** may be illuminated when the electrical prongs on an electrical plug extend into the receptacle outlet slots **30**A-**30**C and make an electrical connection with electrical terminals inside of the electrical power outlet **20**. As such, the light ring area **26** provides a light ring which serves as an indication that a plug-in device has been connected to the electrical power outlet and is energized to supply electrical power from the power source.

(29) The electrical power outlet **20** further includes a plurality of electrical terminals **40**A-**40**C which are further illustrated in FIG. **3** which are aligned and arranged in the same configuration to the receptacle outlet slots **30**A-**30**C. The electrical terminals **40**A-**40**C are shown mounted on a printed circuit board **52**. The circuit board **52** is generally supported within a rear internal housing bezel **22**B which connects to a front external housing bezel **22**A to form the housing **22**. Each of the electrical terminals **40**A-**40**C is formed having an electrically conductive contacts shaped to receive a corresponding prong from an electrical plug **100**. Disposed between the electrical terminals **40**A-**40**C and the receptacle outlet slots **30**A-**30**C is a support base **42** which generally includes openings **44**A-**44**C. Openings **44**A-**44**C are shaped, sized and aligned with the corresponding slots **30**A-**30**C and corresponding terminals **40**A-**40**C to allow the prongs on the electrical plug to extend through slots **30**A-**30**C, and through openings **44**A-**44**C and into contact with electrical terminals **40**A-**40**C. As such, the support base **42** helps to support the prongs as they extend into and make contact with electrical terminals **40**A-**40**C.

(30) Also provided on the printed circuit board **52** is a lighting device **48** shown as a light-emitting diode (LED). The lighting device **48** generally extend forward proximate to the faceplate **24** and when activated emits light within the light ring area **26**. As such, an illuminated light ring is provided around the periphery of the faceplate **24**. In addition, a mechanical switch **46** is disposed on the printed circuit board **52** and is aligned with one of the electrical terminals **40**A-**40**C. As such, the switch **46** is aligned to receive and make physical contact with a first prong of the electrical plug when the electrical plug is fully plugged into the electrical power outlet **20**. In the example shown, a first prong **102**B of the electrical plug **100** may extend through slot **30**B, opening **44**B and make contact with a first electrical terminal **40**B. In doing so, the first prong **102**B will forcibly engage the switch **46** and mechanically depress the switch **46** to change state as an indication that an electrical plug is connected to the electrical power outlet **20**. With the switch **46** is engaged by a prong, electrical power is supplied to the electrical terminals **40**A-**40**C to supply power to an electrically-powered device via the electrical plug **100**. In addition, the lighting device **48** is illuminated to illuminate the light ring area **26** to indicate that the electrical power outlet **20** is powered or providing status of the plug depending on the configuration given.

(31) The housing **22** includes both the external front bezel **22**A which connects with the rear internal bezel **22**B. To make the connection, a plurality of flexible fingers **50**B provided on the rear internal bezel **22**B are configured to engage tabs **50**A on the front exterior bezel **22**A to connect and hold the housing assembled together and to contain the parts therein.

(32) With particular reference to FIGS. **4** and **5**, the electrical power outlet **20** is shown having a flexible cover **32** disposed between the plurality of receptacle outlet slots **30**A-**30**C and the plurality of electrical terminals **40**A-**40**C. In the example shown, the flexible cover **32** is located immediately behind the receptacle outlet slots **30**A-**30**C formed in the faceplate **24** and forward of the support base **42**. The flexible cover **32** is a flexible thin membrane that includes a plurality of slits **34**A-**34**C formed in the regions immediately proximate the corresponding receptacle outlet slots **30**A-**30**C. Each of the slits **34**A-**34**C allows the thin membrane to flex and separate to an open position as a prong of the electrical plug is forcibly inserted through the corresponding receptacle outlet slots **30**A-**30**C and against the flexible cover **32**. When this occurs, the slits **34**A-**34**C open to

allow the corresponding prongs to extend therethrough and to make contact with the corresponding electrical terminals **40**A-**40**C. When the prongs of the electrical plug are removed from the electrical power outlet **20**, the edges of the slits **34**A-**34**C of the flexible cover **32** return with memory to the closed position. This allows for the blockage of particles such as dirt and water or any unwanted objects from tampering with or otherwise entering the electrical power outlet **20**.

(33) The flexible cover **32** may be made of a thin flexible material which may include a polymeric material such as plastic or rubber, for example, and has a sufficiently thin thickness to allow the membrane to flex between open and closed positions during insertion or removal of the plug. In the example shown, the slits **34**A-**34**C generally include an I-shape slit with an elongated vertical slit extending between upper and lower horizontal sits are seen in FIG. **5**. Slit **34**A further includes an extra horizontal slit extending midway between the upper and lower slits. However, it should be appreciated that other shapes and sizes of slits **34**A-**34**C may be provided in the flexible cover **32**.

(34) With reference to FIGS. **6** and **6**A, one example of an electrical plug **100** positioned to engage the electrical power outlet **20** is shown in FIG. **6**. As seen in FIG. **6**A, the electrical plug **100** is fully inserted into the electrical outlet **20** such that the prongs **102**A-**102**C contact the corresponding terminals **40**A-**40**C. In doing so, first prong **102**B makes contact with switch **46** and depresses the switch **46** so as to change the state of the switch **46** to activate electric power to the light source **48** and to activate electric power supplied to the electrical plug **100** to power an external electrically-powered device connected thereto.

(35) Referring to FIGS. **7-10**, an electrical power outlet **20**′ is illustrated, according to a second embodiment. The electrical power outlet **20**′ may likewise be located on a motor vehicle. In this embodiment, the electrical power outlet **20**′ includes a similar arrangement of a faceplate **24** having receptacle outlet slots **30**A-**30**C arranged to receive an electrical plug to power an electric powered device. Electrical terminals **40**A-**40**C are mounted on a printed circuit board **52** and a switch is aligned with the first electrical terminal **40**B. Located between the faceplate **24** and the plurality of electrical terminals **40**A-**40**C is the flexible cover **32** which includes the slits **34**A-**34**C.

(36) In this embodiment, a light transparent optical ring **60** is disposed proximate the light ring area **26** to provide a luminescent illumination in the light ring area **26** extending around the faceplate **24**. The light source **48** is aligned to backlight the light transparent optical ring **60**. The optical ring **60** may include an optically transparent material such as diffuse light pipe that illuminates a ring of light when illuminated by a light source **48** onto a receiver **54** on the back side of the optical ring **60**. The optical material may include glass having diffuse light properties.

(37) Referring to FIGS. **11** and **12**, an electrical power outlet **20**″ is illustrated according to a further embodiment. In this embodiment, the electrical power outlet **20**″ includes a similar arrangement of components except the flexible cover **32** is disposed behind slot **30**A and internal tabs **35**B and **35**C are positioned behind slots **30**B and **30**C. The tabs **35**B and **35**C are made up of two flexible arms that normally abut and are biased towards each other in a closed position and flex to an open position when a prong is forcibly extended therethrough. As such, tabs **35**B and **35**C may operate as a flexible cover.

(38) Accordingly, the electrical power outlet **20** advantageously provides for a flexible cover **32** disposed between the plurality of receptacle outlet slots and the plurality of terminals and arranged with a plurality of slits configured to receive the plurality of prongs of the electrical plug for covering the plurality of terminals to prevent contaminants from entering the electrical power outlet **20** and tampering of the electrical power outlet **20**.

(39) It is to be understood that variations and modifications can be made on the aforementioned structure without departing from the concepts of the present disclosure, and further it is to be understood that such concepts are intended to be covered by the following claims unless these claims by their language expressly state otherwise.

Claims

1. An electrical power outlet comprising: a housing; a faceplate having a plurality of slots configured to receive a plurality of prongs on an electrical plug; a plurality of electrical terminals configured to receive the plurality of prongs to provide electrical connections; a flexible cover disposed between the plurality of slots and the plurality of terminals, the flexible cover having a plurality of slits configured to receive the plurality of prongs in an open position and for covering access to the plurality of terminals in a closed position; a switch coupled to at least one of the plurality of electrical terminals to detect the connection of a first prong to a first electrical terminal of the plurality of electrical terminals, wherein the switch activates electrical power supplied to the plurality of prongs when the switch detects connection of the first prong to the first electrical terminal; and a circuit board, wherein the plurality of electrical terminals are mounted on the circuit board and the switch is aligned to be engaged with the first electrical terminal.
2. The electrical power outlet of claim 1, wherein the switch is coupled to the circuit board.
3. The electrical power outlet of claim 1, wherein the cover comprises a membrane of flexible material.
4. The electrical power outlet of claim 1, wherein the housing comprises a first bezel configured to connect to a second bezel.
5. The electrical power outlet of claim 1, wherein the electrical power outlet is located on a motor vehicle.
6. The electrical power outlet of claim 1 further comprising a light source configured to illuminate at least a region proximate the faceplate when the switch detects connection of the first prong to the first electrical terminal.
7. The electrical power outlet of claim 6, wherein the light source comprises a light transmissive optic element forming a ring surrounding the faceplate.
8. The electrical power outlet of claim 6, wherein the light source comprises a light-emitting diode (LED).
9. An electrical power outlet comprising: a housing; a faceplate having a plurality of slots configured to receive a plurality of prongs on an electrical plug; a plurality of electrical terminals configured to receive the plurality of prongs to provide electrical connections; a flexible cover disposed between the plurality of slots and the plurality of terminals, the flexible cover having a plurality of slits configured to receive the plurality of prongs in an open position and for covering access to the plurality of terminals in a closed position; a switch coupled to a first electrical terminal of the plurality of electrical terminals to detect the connection of a first prong to the first electrical terminals, wherein the switch activates electrical power supplied to the plurality of prongs when the switch detects connection of the first prong to the first electrical terminal; a circuit board, wherein the plurality of electrical terminals are mounted on the circuit board and the switch is aligned to be engaged with the first electrical terminal, wherein the switch is coupled to the circuit board; and a light source configured to illuminate a region proximate the faceplate when the switch detects connection of the first prong to the first electrical terminal.
10. The vehicle of claim 9, wherein the cover comprises a membrane of flexible material.
11. The vehicle of claim 9, wherein the plurality of slits includes I-shaped slits.
12. A vehicle comprising: an electrical power source; power lines receiving electrical power from the electrical power source; and an electrical power outlet coupled to the power lines, the electrical power outlet comprising: a housing; a faceplate having a plurality of slots configured to receive a plurality of prongs on an electrical plug; a plurality of electrical terminals configured to receive the plurality of prongs to provide electrical connections; a flexible cover disposed between the plurality of slots and the plurality of terminals, the flexible cover having a plurality of slits configured to receive the plurality of prongs in an open position and for covering access to the plurality of

terminals in a closed position; a switch coupled to a first electrical terminal of the plurality of terminals to detect the connection of a prong in the first electrical terminal, wherein the switch activates electrical power supplied to the plurality of prongs when the switch detects connection of the first prong to the first electrical terminal; and a circuit board, wherein the plurality of electrical terminals and the switch are mounted on the circuit board and the switch is aligned to be engaged with the at the first electrical terminal on the circuit board.
13. The vehicle of claim 12 further comprising a light source configured to illuminate at least a region proximate the faceplate when the switch detects connection of the first prong to the first electrical terminal or change color according to a status of the plug.
14. The vehicle of claim 13, wherein the light source comprises a light transmissive optic element forming a ring surrounding the faceplate.
15. The vehicle of claim 13, wherein the light source comprises a light-emitting diode (LED).
